

WALMART SALES PERFORMANCE DASHBOARD

USING MICROSOFT POWER BI

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TOOL USED : MICROSOFT POWER BI DESKTOP

1. Introduction

For this project, I created a **Walmart Sales Performance Dashboard** using Microsoft Power BI. I used the Walmart weekly sales dataset to analyse sales performance across different stores and time periods.

The dataset includes store number, weekly sales, date, fuel price, CPI, unemployment rate, and holiday information. Using Power BI charts, KPI cards, and slicers, I converted the raw data into an easy-to-understand interactive dashboard.

2. Objectives

- To analyse Walmart's weekly sales.
- To compare sales across different stores.
- To compare holiday and non-holiday sales.
- To identify sales trends over time.
- To check the relationship between fuel price and sales.
- To create an interactive Power BI dashboard.
- To present data in a simple visual format.

3. Dataset Used

The **Walmart Sales Dataset** contains weekly sales information from different Walmart stores.

- **Store** – Store number.
- **Date** – Sales date.
- **Weekly Sales** – Weekly sales amount.
- **Fuel Price** – Fuel price during the period.
- **CPI** – Consumer Price Index.
- **Unemployment** – Unemployment rate.
- **Holiday Flag** – Shows whether the week was a holiday or not.

4. Data Preparation

I imported the Walmart Excel dataset into Power BI Desktop and checked the data formats.

The **Date** column was converted into the correct date format. Numerical columns such as Weekly Sales, Fuel Price, CPI, and Unemployment were also checked and formatted properly.

After preparing the data, I created DAX measures for the dashboard.

5. DAX Measures

The main DAX measures created were:

- **Total Sales** – Calculates overall sales.
- **Average Sales** – Calculates average weekly sales.
- **Store Count** – Shows the number of stores.
- **Maximum Sales** – Shows the highest weekly sales.

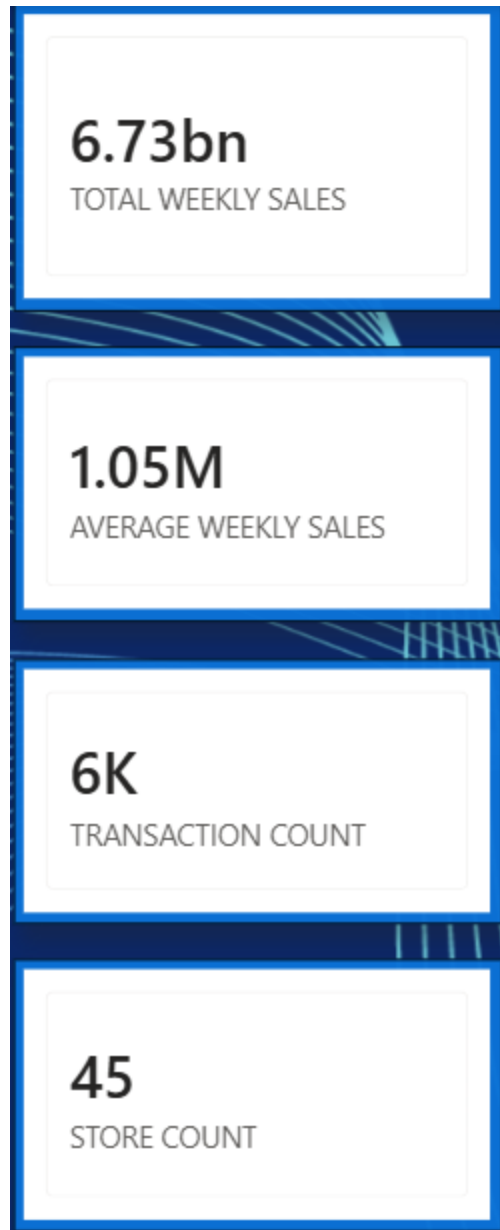
These measures were used in the KPI cards.

6. Visualizations Created

KPI Cards

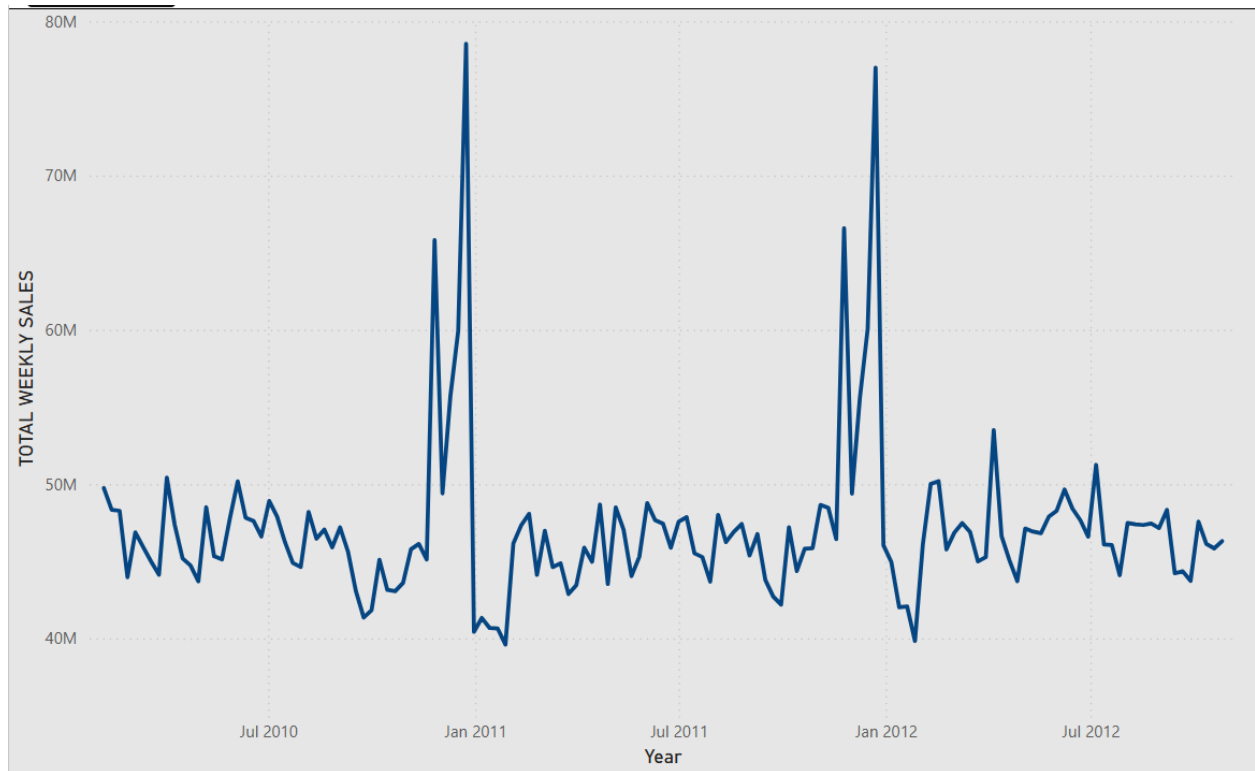
Four KPI cards were created to show:

- Total Sales
- Average Sales
- Number of Stores
- Maximum Weekly Sales



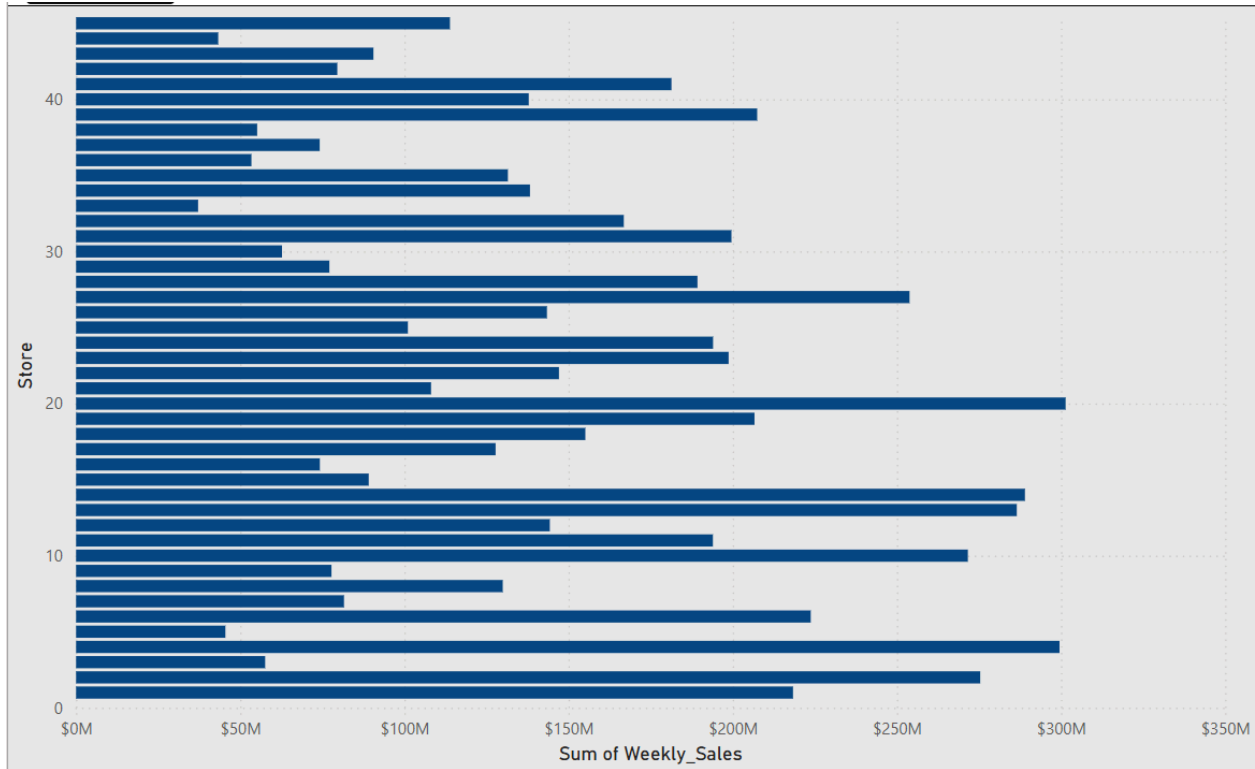
Weekly Sales Trend

A line chart was used to show changes in weekly sales over time.



Sales by Store

A bar chart was used to compare sales performance between different stores



Holiday vs Non-Holiday Sales

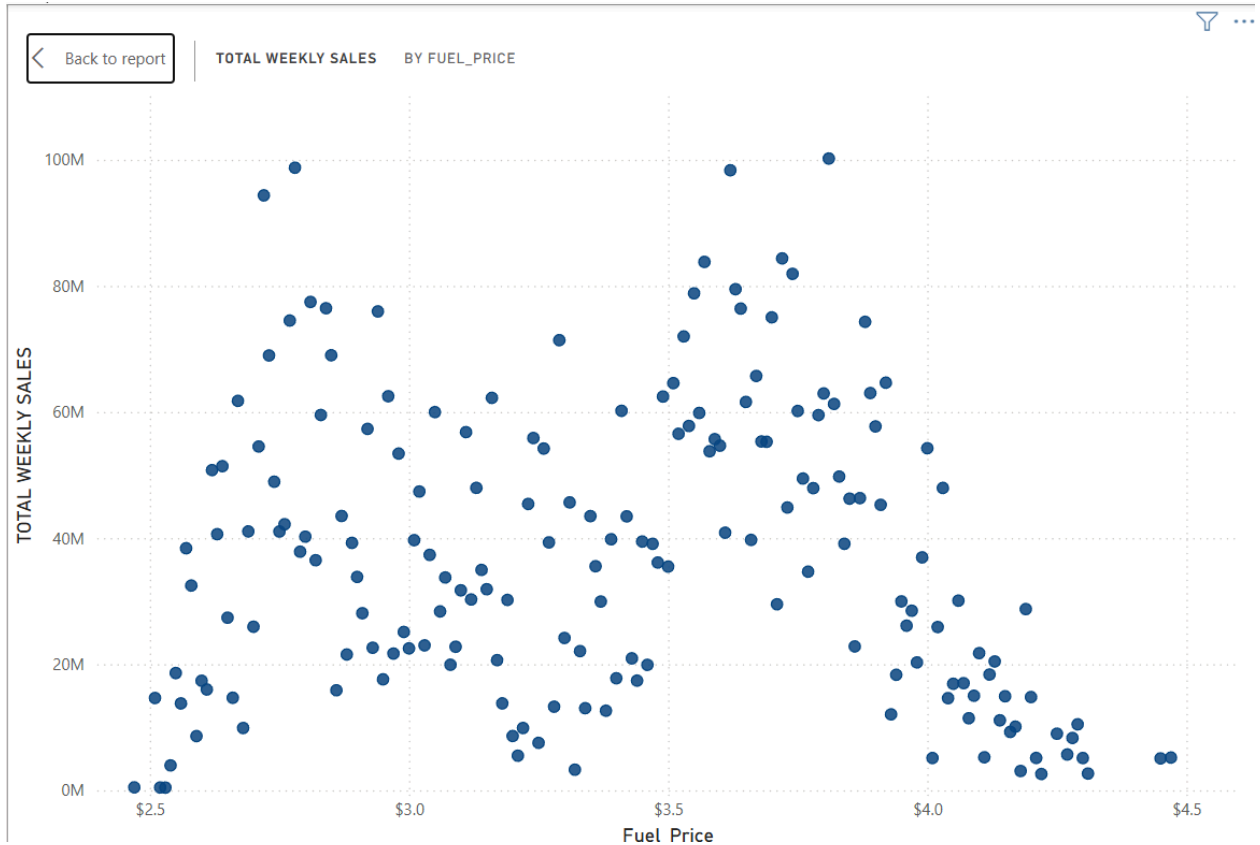
A pie chart was used to compare sales during holiday and normal weeks.

Sales Contribution by Store

A treemap was used to show each store's contribution to total sales.

Fuel Price and Weekly Sales

A scatter chart was used to check the relationship between fuel price and weekly sales. No strong relationship was observed.



CPI and Unemployment Trends

A line chart was used to observe CPI and unemployment changes over time.

7. Slicers Used

Three slicers were added to make the dashboard interactive:

- **Year**
- **Store**
- **Holiday Status**

FILTERS

STORE

All

DATE

2/5/201010/26/2012

HOLIDAY FLAG

Holiday_Flag

☐ Select all

☒ HOLIDAY

☒ NON HOLIDAY

8. Key Findings

The main findings from the dashboard were:

- Total sales were approximately **\$6.73 billion**.
- Average weekly sales were around **\$1.05 million**.
- The dataset included **45 stores**.
- Highest weekly sales were approximately **\$3.82 million**.
- Sales performance differed across stores.

- Non-holiday weeks had higher total sales mainly because there were more non-holiday weeks.
- Weekly sales changed during different periods.
- Fuel prices did not show a strong relationship with weekly sales.

9. Conclusion

This project helped me understand how to use **Power BI for business data analysis**. I learned how to prepare data, create DAX measures, design charts, and use slicers.

The dashboard made it easier to understand Walmart's sales performance and identify important trends. Overall, this project improved my practical knowledge of **Power BI and data visualization**.